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Phase aberration separation for holographic microscopy by alternating direction sparse optimization

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Abstract: The morphology and dynamics of label-free tissues can be exploited by sample-induced changes in the optical field from quantitative phase imaging. Its sensitivity to subtle changes in the optical field makes the reconstructed phase susceptible to phase aberrations. We import variable sparse splitting framework on quantitative phase aberration extraction based on alternating direction aberration free method. The optimization and regularization in the reconstructed phase are decomposed into object terms and aberration terms. By formulating the aberration extraction as a convex quadratic problem, the background phase aberration can be fast and directly decomposed with the specific complete basis functions such as Zernike or standard polynomials. Faithful phase reconstruction can be obtained by eliminating global background phase aberration. The aberration-free two-dimensional and three-dimensional imaging experiments are demonstrated, showing the relaxation of the strict alignment requirements for the holographic microscopes.

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1. Introduction

Rapid and accurate estimation of the viability of biological cells is important for assessing the impact of drugs, physical or chemical stimulants. Assessing cell viability often requires mixing cell populations with reagents to convert substrates to colored or fluorescent products [1]. Digital holographic microscopy (DHM) reconstructs the entire complex wavefront, including both the amplitude and phase of the light field [2], becoming a powerful technique of label-free quantitative phase imaging. It combines properties from microscopy, holography, and light scattering techniques for the morphology and dynamics non-destructive imaging of fully transparent structures based on quantitative intrinsic contrast [3].

Optical aberration limits the ability to reveal structures at diffraction-limited resolutions and it may cause inconsistencies between images obtained in different setups. By measuring the complex amplitude of light, DHM provides quantitative phase reconstruction of specimens [4–7]. Its sensitivity to subtle changes in the optical field makes the phase reconstruction susceptible to optical aberrations. By introducing an off-axis reference wave, the incident object beam and reference beam produce a constant tilt aberration. Optical aberration in holographic imaging depends on the sample or sample holders because of the mismatch in refractive index (RI) between optical elements and inaccurate estimation of the reference wave [8,9]. The use of microscopic objective (MO) introduces phase aberrations that can be superposed over the biological sample. Besides, high-order aberrations such as astigmatism and coma may also exist in the final image due to the imperfect construction of the system and alignment of components. A successful image reconstruction theoretically requires tedious alignment and precise measurement of the system parameters, such as the reconstruction distance, the incident angle of the reference beam, and the focal length of the MO, which is often hard to achieve for industrial applications. These phase aberrations may severely hinder the quantitative measurement of the object's thickness or height with distorted visualization [10].

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To eliminate phase aberration, two categories are considered: physics compensation in optical recording and numerical compensation in numerical reconstruction. Physical compensation methods are conducted by introducing the same wavefront curvature in the reference beam, such as introducing either the same objective lens [11–13] or by adopting an electrically tunable lens configuration [14–17]. The tedious alignment and precise measurement of the configuration are also required. Another compensation method in this category is double exposure [18,19]. A reference hologram needs to be recorded without the presence of the specimen. The phase aberration may drift with the changes of focus and other experimental conditions especially long-term imaging. Another category based on the total single phase of the object including aberration is numerical compensation. Fitting-based numerical methods have been proposed by using pre-defined aberration models and least-square fitting with the spherical surface [20], parabolic function [21–23], the standard polynomials [24,25], or Zernike polynomials [26–31]. The estimate of the phase mask is prone to be deviated by the phase values of areas occupied by the sample. The linear and spherical phases without high-order aberration can also be directly extracted by using principal component analysis (PCA) [23,32,33]. Without pre-defined aberration models, the low-frequency slight phase aberrations can also be estimated from low-pass spatial filtering [34]. By collecting many phase fields under different angles, pupil aberrations can also be eliminated by multiple-frame detections based on cross-correlation operation between each coherent aperture [35,36]. Relying on the improvement of computing, the learning-based compensation method can automatically detect the specimen-free background [29,37,38]. The sample-free region is detected based on a convolutional neural network and it allows for Zernike polynomial fitting to obtain a phase mask that is undisturbed by the sample phase. By integrating the compact optical-electronic module with diffractive neural networks printed on imaging sensors [39], the Zernike-based pupil phase distributions can be directly reconstructed from an incident point spread function.

The background phase usually keeps the characteristic of moderate. It can be considered that the systematic disturbance with a known model is superimposed on an unknown object function. We propose a phase variable splitting framework for aberration extraction based on alternating direction sparse optimization, named alternating direction aberration-free (ADAF) method. Different from the least-square for the entire measured field, the optimization and regularization in phase are simultaneously decomposed into two solutions with object terms and aberration terms. By formulating the aberration extraction as an inverse problem, an accurate aberration surface is separated from the total reconstructed phase by imposing sparsity regularization. The background aberration can be directly decomposed with the Zernike or standard polynomials based on the ADAF method. Faithful phase reconstruction can be obtained which can relax the strict alignment requirements for tolerating system aberrations. Different samples such as phase resolution target, label-free normal colon tissue slice, label-free normal colonic mucosa, and micro-beads can be visualized experimentally with eliminating aberration.

2. Methodology

The schematic of the experimental holographic phase microscopy based on Mach–Zehnder interferometer with the transmissive mode is shown in Fig. 1(a). The light source is Nd: YAG laser at the wavelength of 532 nm with fiber-coupled-out mode. The source is collimated by the lens (L) with a quasi-plane wave and it is divided into an illumination arm and a reference arm by a beam splitter (BS) 1. The L and the condenser lens (CL) are combined into a 4f system and the sample (S) is illuminated. The mirror (M) is used to adjust the reflection angle. The MO and the tube lens (TL) are combined for the transmissive microscopy. The L3 and L4 are combined into a magnification 4f system. The object wave and the reference wave are interfered by introducing BS2. The polarizer (P) is placed in front of the camera. The object wave output from the microscopy interfered with the reference wave in fringe patterns, which are recorded

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by a complementary metal–oxide–semiconductor (CMOS) camera. The hologram is shown in Fig. 1(b). Its intensity represents the interference:

$$I_H(\mathbf{r}) = |R(\mathbf{r})|^2 + |S(\mathbf{r})|^2 + S(\mathbf{r})R^*(\mathbf{r}) + S^*(\mathbf{r})R(\mathbf{r}), \tag{1}$$

where I_H is the intensity, S is the complex function of the object, R is the complex function of the reference wave, * denotes the complex conjugate. $\mathbf{r}$ is the transverse space vector with $\mathbf{r} = (x, y)$. Figure 1(c) shows the spectrum of hologram. The $|R|^2 + |S|^2$ is the autocorrelation (AC) term which is considered as the zeroth order of diffraction. SR^* and S^*R describe the sample and its conjugate term with the 1st order term and the -1st order term, respectively. The hologram is Fourier transformed and filtered with pupil function $F(\mathbf{r})$:

$$\begin{aligned} O_{filter}(\mathbf{r}) &= \mathcal{F}^{-1}\{\mathcal{F}[I_H(\mathbf{r})] \cdot F(\mathbf{k})\} \\ &= R_0|S(\mathbf{r})|e^{i[\varphi_s(\mathbf{r})+\varphi_a(\mathbf{r})+\mathbf{k}_R\cdot\mathbf{r}]}, \end{aligned} \tag{2}$$

where $O_{filter}(\mathbf{r})$ is the complex amplitude of the filtering hologram. $\mathbf{k}$ is the transverse frequency vector. $\mathcal{F}$ and $\mathcal{F}^{-1}$ denotes the Fourier transform and inverse Fourier transform, respectively. R_0 is the amplitude of the reference wave. $\varphi_s(\mathbf{r})$ is the phase distribution of sample and $\varphi_a(\mathbf{r})$ is the phase distribution of aberration. $\mathbf{k}_R$ represents the two-dimensional (2D) spatial carrier frequency caused by off-axis holography. The phase of $O_{filter}(\mathbf{r})$ is shown in Fig. 1(d). The unwrapped phase could be written in this form as $\varphi(\mathbf{r}) = \varphi_s(\mathbf{r}) + \varphi_a(\mathbf{r})$, as shown in Fig. 1(e), where $\varphi_s(\mathbf{r})$ is the sample phase, $\varphi_a(\mathbf{r})$ is the background aberration introduced by the optical elements.

The purpose of numerical aberration compensation is to find an approximation that is close to the real aberration [25]. In order to approximate the aberration, a theoretical fitting model is desired. Optical aberration in holographic imaging depends on the sample or sample holders because of the mismatch in RI between optical elements and inaccurate estimation of the reference wave [8,9]. The base of the aberration can be defined by standard polynomials or Zernike polynomials [26]:

$$\varphi_{as}(\mathbf{r}) = -\sum_{\alpha=\beta=0}^{\alpha+\beta=m} P_{\alpha\beta}x^\alpha y^\beta, \tag{3}$$

$$\varphi_{az}(\mathbf{r}) = -\sum_{\gamma=0}^{n} P_\gamma Z_\gamma(\mathbf{r}), \tag{4}$$

where $P_{\alpha\beta}$ and P_γ are the coefficient of each order in standard polynomials and Zernike polynomials, respectively. The α and β are order of the coordinates x and y in different directions. The m and n are the order of polynomials. Z_γ is the γth order Zernike polynomial. The aberration measurement can be modeled as a linear combination of the Zernike coefficient and polynomial. Figure 1(f) displays only the first 15 Zernike basis functions. We represent it as the matrix–vector multiplication: $\mathbf{\Phi} = \mathbf{Ap}$, where $\mathbf{\Phi}$ is the vector containing the phase aberration, $\mathbf{p} = \begin{pmatrix} p_0, p_1, p_2 \dots p_n \end{pmatrix}^T$ denotes a vector representing the coefficient, and $\mathbf{A} = \begin{pmatrix} \mathbf{a}_0, \mathbf{a}_1, \mathbf{a}_2 \dots \mathbf{a}_n \end{pmatrix}$ is the matrix of the polynomial base, and $\mathbf{a}_n$ is the polynomial vector, and n is the number of orders. $\mathbf{\Phi}$ and $\mathbf{a}_n$ are $N_xN_y \times 1$ vector if the image contains $N_x \times N_y$ elements. Finding the coefficient is a typical ill-posed inverse problem. Conventionally, this problem is solved by the least squares method. The residual between the ground truth and the predicted value by the model is minimized with the ℓ_2-norm loss function. In order to introduce the ADAF method, we separate target object and phase aberration, and they can be solved by a

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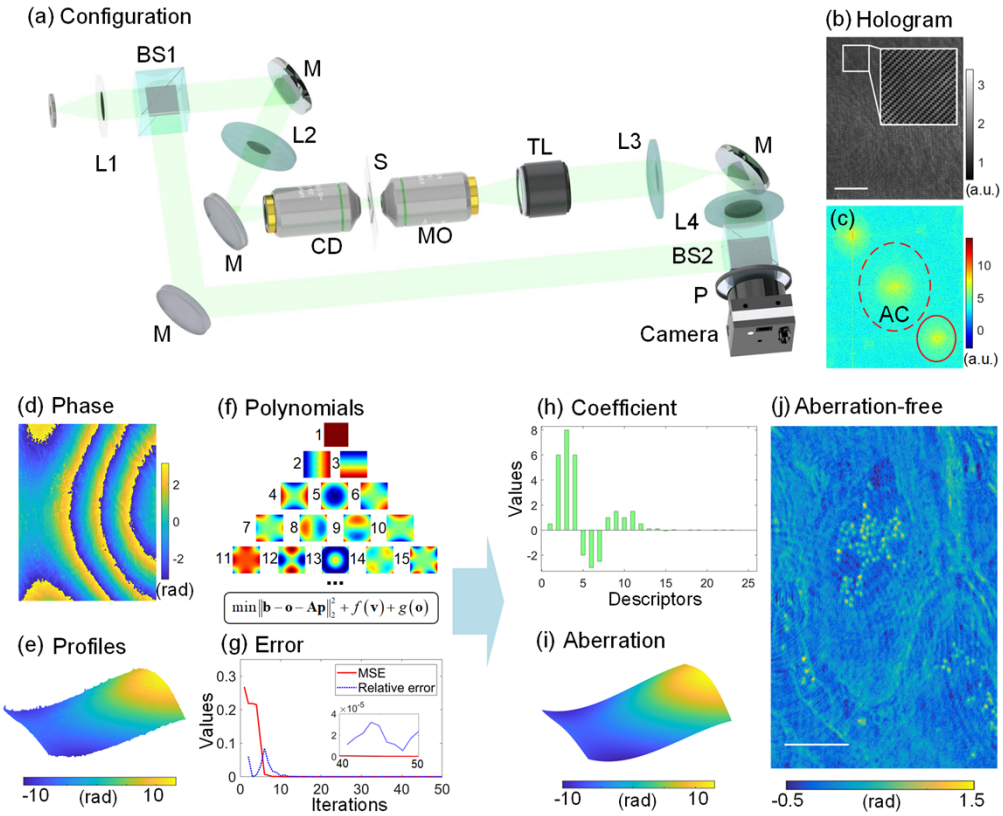


Fig. 1. (a) Schematic setup of transmissive holographic phase microscopy. L: lens; BS: Beam splitter; M: Mirrors; CL: Condenser lens; MO: Microscope objective; TL: Tube lens; P: Polarizer. (b) Raw hologram from holographic phase microscopy in label-free normal colon tissue slice. (c) The spectrum of the hologram (b). (d) Reconstructed phase from the 1st order in the spectrum (c). (e) Pseudo 3D view of unwrapped phase in (d). (f) Visualization of the first 15 Zernike basis polynomials. (g) The reconstructed MSE and relative error with different numbers of iterations. (h) The final reconstructed coefficients in each descriptor. (i) The aberration term by constructing the profiles with reconstructed coefficients and basis functions. (j) Aberration-free phase from (e) by eliminating aberration (i). The scale bar is 30 µm.

minimization problem:

$$\begin{aligned} &\min||\mathbf{b}-\mathbf{o}-\mathbf{A}\mathbf{p}||_2^2+f(\mathbf{v})+g(\mathbf{o}),\\ &\text{s.t. } \mathbf{v}=\mathbf{p},\ \mathbf{A}\mathbf{p}+\mathbf{o}=\mathbf{b}, \end{aligned} \tag{5}$$

where $\mathbf{b}$ is the total phase reconstruction which describes the phase induced by the sample and the imaging system, $\mathbf{o}$ is assumed as the sample distribution under a certain angle of illumination, $\mathbf{v}$ is the Zernike coefficients in each polynomial. The $f(\mathbf{v})$ and $g(\mathbf{o})$ are objective functions with $f(\mathbf{v})=\mu_1||\mathbf{v}||_1$ and $g(\mathbf{o})=\mu_2||\mathbf{o}||_1$, where μ_1 and μ_2 are parameters. The alternating direction optimization is applicable for this given variable splitting strategy of Eq. (5) [40,41]. The corresponding augmented Lagrangian problem is:

$$\min_{\mathbf{p},\mathbf{v},\mathbf{o}} \mu_1||\mathbf{v}||_1+\mu_2||\mathbf{o}||_1+\lambda_1^T(\mathbf{v}-\mathbf{p})+\frac{\rho_1}{2}||\mathbf{v}-\mathbf{p}||_2^2+\lambda_2^T(\mathbf{A}\mathbf{p}+\mathbf{o}-\mathbf{b})+\frac{\rho_2}{2}||\mathbf{A}\mathbf{p}+\mathbf{o}-\mathbf{b}||_2^2, \tag{6}$$

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where $\lambda_1 \in \mathbb{R}^n$, $\lambda_2 \in \mathbb{R}^{N_x x N_y}$ are multipliers, and $\rho_1, \rho_2 > 0$ are penalty parameters. Then, the alternating direction method is applied to minimize the augmented Lagrangian problem in Eq. (6) with respect to **v**, **p,** and **o** alternately. The **p**-subproblem, namely minimizing Eq. (6) with respect to **p**, is given by

$$\begin{aligned} &\min_{\mathbf{p}} -\lambda_1^T \mathbf{p} + \tfrac{\rho_1}{2}||\mathbf{v}-\mathbf{p}||_2^2 + \lambda_2^T \mathbf{A}\mathbf{p} + \tfrac{\rho_2}{2}||\mathbf{A}\mathbf{p}+\mathbf{o}-\mathbf{b}||_2^2, \\ &\Leftrightarrow \min_{\mathbf{p}} \tfrac{1}{2}\mathbf{p}^T(\rho_1\mathbf{I}+\rho_2\mathbf{A}^T\mathbf{A})\mathbf{p} - (\rho_1\mathbf{v}+\lambda_1+\rho_2\mathbf{A}^T\mathbf{b}-\rho_2\mathbf{A}^T\mathbf{o}-\mathbf{A}^T\lambda_2)^T\mathbf{p}. \end{aligned} \tag{7}$$

Note that it is a convex quadratic problem [40]. Hence, it reduces to solving the following linear system:

$$(\rho_1\mathbf{I}+\rho_2\mathbf{A}^T\mathbf{A})\mathbf{p} = \rho_1\mathbf{v}+\lambda_1+\rho_2\mathbf{A}^T\mathbf{b}-\rho_2\mathbf{A}^T\mathbf{o}-\mathbf{A}^T\lambda_2. \tag{8}$$

Minimizing Eq. (6) with respect to **v** gives the following **v**-subproblem:

$$\min_{\mathbf{v}} \mu_1||\mathbf{v}||_1 + \lambda_1^T\mathbf{v} + \frac{\rho_1}{2}||\mathbf{v}-\mathbf{p}||_2^2, \tag{9}$$

which has a closed-form solution by the one-dimensional shrinkage or soft thresholding,

$$\mathbf{v} = \frac{\left(\mathbf{p}-\frac{\lambda_1}{\rho_1}\right)}{\left|\mathbf{p}-\frac{\lambda_1}{\rho_1}\right|} \max\left\{\left|\mathbf{p}-\frac{\lambda_1}{\rho_1}\right| - \frac{\mu_1}{\rho_1}, 0\right\}. \tag{10}$$

With the same principle, minimizing Eq. (6) with respect to **o** gives the following solution,

$$\mathbf{o} = \frac{\left(\mathbf{b}-\mathbf{A}\mathbf{p}-\frac{\lambda_2}{\rho_2}\right)}{\left|\mathbf{b}-\mathbf{A}\mathbf{p}-\frac{\lambda_2}{\rho_2}\right|} \max\left\{\left|\mathbf{b}-\mathbf{A}\mathbf{p}-\frac{\lambda_2}{\rho_2}\right| - \frac{\mu_2}{\rho_2}, 0\right\}. \tag{11}$$

Finally, the multipliers λ_1 and λ_2 are updated in the standard way [40],

$$\begin{cases} \lambda_1 \leftarrow \lambda_1 - \tau_1\rho_1(\mathbf{p}-\mathbf{v}) \\ \lambda_2 \leftarrow \lambda_2 - \tau_2\rho_2(\mathbf{b}-\mathbf{o}-\mathbf{A}\mathbf{p}) \end{cases}, \tag{12}$$

where τ_1, τ_2 are the step lengths. The **v**, **p,** and **o** are iteratively updated by Eqs. (8), (10), (11) and (12), which is the process of ADAF method. The convergence is guaranteed by the existing alternating direction theory [41]. For $\rho_1, \rho_2 > 0$ and $\tau_1, \tau_2 \in \left(0, \left(\sqrt{5}+1\right)/2\right)$, the sequence $(\mathbf{o}^{(k)}, \mathbf{p}^{(k)})$ generated by the algorithm from any initial point $(\mathbf{o}^{(0)}, \mathbf{p}^{(0)})$ converges to $(\mathbf{o}^*, \mathbf{p}^*)$. The conventional fitting method can be considered the least square method between the aberration model and the total phase. The least squares method for regression analysis is to minimize an l_2-norm loss function. It is equal to minimizing the amount of energy in the system and is mathematically simple to solve. But the poor fitting results may be produced for which the unknown coefficients have nonzero energy, and the estimation is prone to be deviated by the phase values of areas occupied by the sample. The l_1-norm can be turned to the well-known compressed sensing theory. To enforce the sparsity constraint when solving this linear equation, the sparse solution is feasible to constraints given by the encoding matrix of aberration and the observation object **o** with highly accurate and efficient [42,43]. Figure 1(g) shows the convergence of the ADAF method by using dual regularization in aberrations extraction. The red solid line shows the reconstructed mean square errors (MSE, $E_s = \left[\sum_{ii}\sum_{jj}|r_s(ii,jj) - r_0(ii,jj)|^2\right]/(N_xN_y)$, where

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$r_s(ii,jj)$ is the estimated distribution and $r_0(ii,jj)$ is the initial distribution) with a number of iterations in dual regularization. The reconstructed MSE reaches the order 10^{-4} by using the ADAF method with dual regularization before 10 iterations. The blue dotted line shows the relative error between adjacent iterations. The relative error between the adjacent iterations is lower than 10^{-4} after 10 iterations. Figure 1(h) shows the reconstructed coefficients $\mathbf{p}^*$ in each descriptor. By constructing the profiles with coefficients and basic functions, the aberration $\varphi_a(\mathbf{r})$ can be reconstructed from the total phase $\varphi(\mathbf{r})$ in Fig. 1(e), as the pseudo three-dimensional (3D) view shown in Fig. 1(i). The sample phase $\varphi_s(\mathbf{r})$ in Fig. 1(j) is obtained by subtracting the aberration $\varphi_a(\mathbf{r})$ from $\varphi(\mathbf{r})$. The ADAF method decomposed the single total phase from DHM into two solutions with object terms and aberration terms. By formulating the aberration extraction as a convex quadratic problem, the aberration terms can be described by a set of specific complete basis functions. An accurate aberration surface is separated from the total reconstructed phase by imposing the sparsity regularization on the object and the coefficients of bases. It returns the reconstructed phase to the distribution with a background 0 rad reference value.

3. Results and discussion

3.1. Numerical calculation analysis

Figure 2 shows the numerical calculation results of the aberration detection. Figures 2(a1) and 2(a2) present 2D phase distribution and its corresponding pseudo 3D view, respectively. The phase field is defined with 1000×1000 pixels and the corresponding area is 3.5×3.5 mm^2. The aberrations are considered both standard polynomials and Zernike polynomials. The step lengths τ_1, τ_2 are both set the $\left(\sqrt{5}+1\right)/2$ which is the upper bound for theoretical convergence guarantee [40]. The ρ_1, ρ_2 are set to near the value of 0.01 and 0.1, respectively. The parameters μ_1, μ_2 are set to 0.01 and 1, respectively. Figures 2(b1)–2(b3) are the aberration produced using the standard polynomials from the 1st to the 2nd, the 3rd, and the 4th order, respectively. It produces $(m^2 + 3m + 2)/2$ terms of polynomials if the number order is m. Figures 2(c1)–2(c3) are the wrapped phase fields with aberrations and Figs. 2(d1)–2(d3) are the corresponding pseudo 3D view of unwrapped phase, respectively. The reconstructed aberrations by using ADAF method are shown in Figs. 2(e1)–2(e3), and the reconstructed phases after aberrations compensations are shown in Figs. 2(f1)–2(f3). The MSE of the reconstructed phase is on the order of 10^{-4} by using the proposed ADAF method. Figures 2(g1)–2(g2) are the reconstructed coefficients of standard polynomials from the 1st term to the 15th term by using the conventional fitting method and the ADAF method, respectively. The black boxes are the original value in each coefficient, while the green and the red boxes are the reconstructed values by using the conventional fitting method and ADAF method, respectively. In order to verify the fitting ability of high-order aberration, we do not deliberately reduce the value of high-order coefficients. The value of each coefficient reveals the accurate aberration extraction of the ADAF method.

Figures 3(a1)–3(a3) are the aberration produced using the Zernike polynomials from the first to the 5th, the 10th, and the 15th term, respectively. The phase field is defined with 1000×1000 pixels and the corresponding area is 1.6×1.6 mm^2. The original phase and the corresponding pseudo-3D view are the same as Figs. 2(a1)–2(a2). Figures 3(b1)–3(b3) are the wrapped phase fields with aberrations and Figs. 3(c1)–3(c3) are the corresponding pseudo 3D view of unwrapped phase, respectively. The reconstructed aberrations by using ADAF method are shown in Figs. 3(d1)–3(d3), and the reconstructed phases after aberrations compensations are shown in Figs. 3(e1)–3(e3). The reconstructed MSE is also on the order of 10^{-4} by using the ADAF method. The accuracy of reconstructed coefficients of Zernike polynomials is also shown in Figs. 3(f1) and 3(f2). The commonly occurred aberrations in holographic microscopy can be fully described by the standard polynomials or Zernike polynomials, such as tilt, astigmatism, and parabolic

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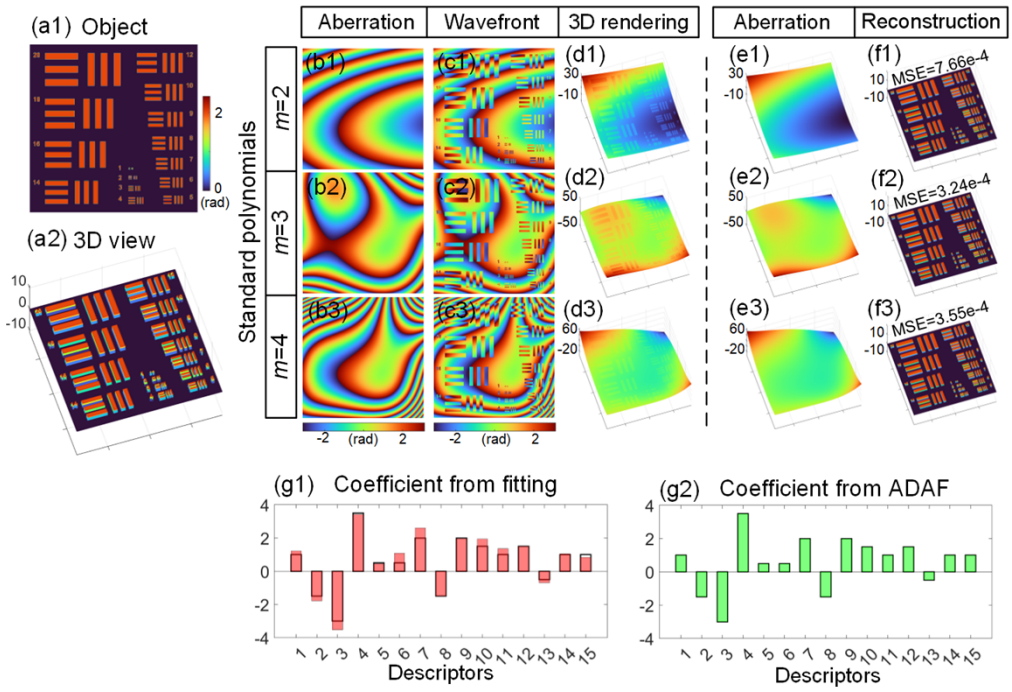


Fig. 2. (a1-a2) Original phase and the corresponding pseudo-3D view. (b1-b3) The aberration was produced using the standard polynomials from the 1st to the 2nd, the 3rd, and the 4th orders, respectively. (c1-c3) The phase fields with aberration from (b1-b3), respectively. (d1-d3) The pseudo-3D view of the unwrapped phase in (c1-c3). (e1-e3) The reconstructed aberrations from (d1-d3). (f1-f3) The reconstructed phases after aberrations compensations from (d1-d3). (g1-g2) The reconstructed coefficients of standard polynomials from the first to the 15th term by the conventional fitting method and the ADAF method, respectively. The black box in each descriptor is the ground truth, the red is the coefficients reconstructed by the conventional fitting method, and the green is the coefficients reconstructed by the ADAF method.

surface. If high-order aberrations have to be removed, the extension of the fitting model to higher order is needed straightforwardly whether the standard polynomials or Zernike polynomials are used. The Zernike polynomials are a better choice in the lens-based holographic imaging system. When the holographic configuration is carefully implemented without obvious deviation, the background phase shows a low spatial frequency fluctuation. The effective coefficients are only concentrated in the first few orders. The 15 orders are enough for reconstructed aberrations. So we choose 15 orders in numerical calculations and experiments.

3.2. *Phase aberration separation in the thin sample*

Figure 4 shows the experimental results of the aberration detection. Figure 4(a) shows the reconstructed wrapped phase of the quantitative phase resolution chart of USAF1951 target with a feature height of 250 nm from the hologram, which can be obtained by performing slightly defocused in the 4*f* system with L3 and L4 in configuration. The numerical aperture (NA) and the magnification are 0.4 and 32×, respectively. The number of pixels is 1200 × 1920 with pixel's size 5.86μm. The linear and spherical phases can also be directly extracted by using PCA and singular value decomposition when the spherical phase aberration dominates the principal components of the background phase [23]. Higher order aberration may not be dominant in the experimental

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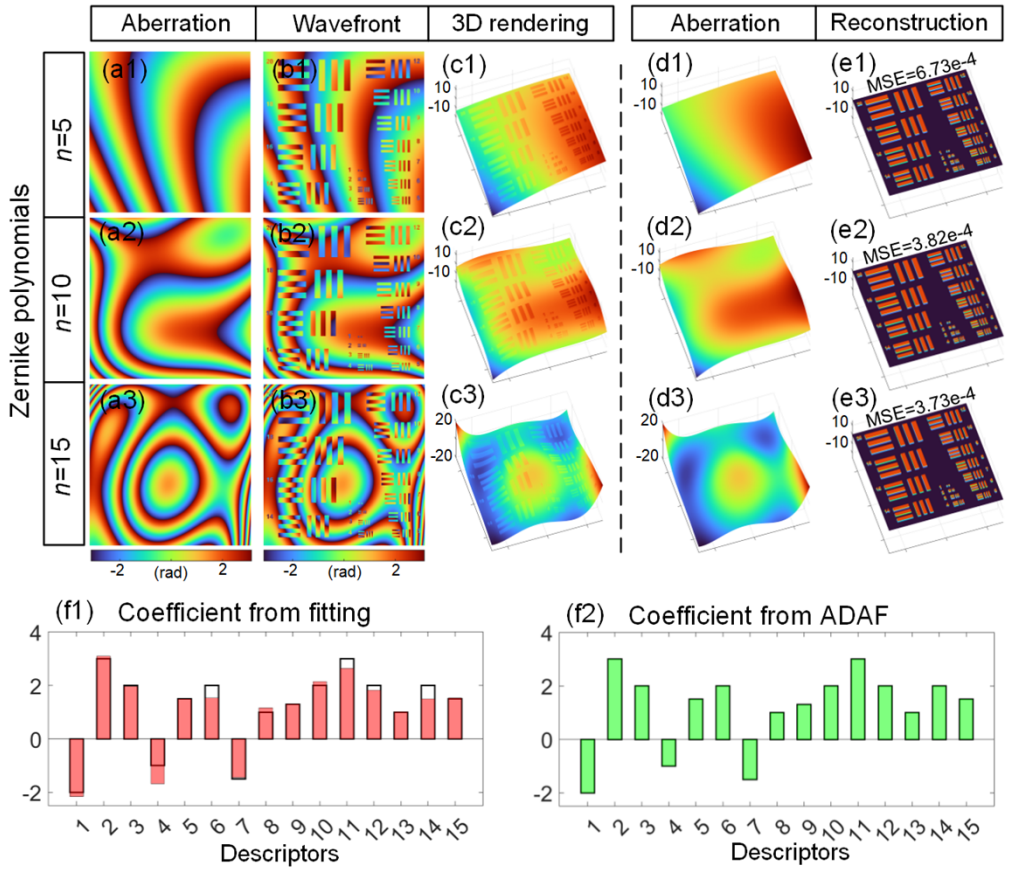


Fig. 3. (a1-a3) The aberration was produced using the Zernike polynomials from the 1st to the 5th, the 10th and the 15th term, respectively. The original phase and the corresponding pseudo-3D view are the same as Figs. 2(a1)-2(a2). (b1-b3) The phase fields with aberration from (a1-a3), respectively. (c1-c3) The pseudo-3D view of the unwrapped phase in (b1-b3). (d1-d3) The reconstructed aberrations from (c1-c3). (e1-e3) The reconstructed phase after aberration compensation from (c1-c3). (f1-f2) The reconstructed Zernike coefficients of polynomials from the first to the 15th by using conventional fitting and ADAF method, respectively. The black box in each descriptor is the ground truth, the red is the coefficients reconstructed by the conventional fitting method, and the green is the coefficients reconstructed by the ADAF method.

reconstruction. Figure 4(b) shows the reconstruction by eliminating PCA. The phase profiles are shown below which reveals that some redundant background phases are kept. Figures 4(c) and 4(d) show the phase compensation using conventional fitting and the ADAF method based on the Zernike polynomials from the 1st term to the 15th term, respectively. Figure 4(e) shows the reconstructed wrapped phase aberration from Fig. 4(a) by using ADAF method. By separating systematic disturbance with known model and object, the sparse object is feasible to constraints given by the encoding matrix, resulting in better phase aberration compensation, while the phase aberration remains visible by using the PCA and fitting method. Figure 4(f) shows the reconstructed phase of the same USAF1951 target but with weaker background aberration than Fig. 4(a). The dominance of spherical phase the has declined by comparing with Fig. 4(a). Figure 4(g) shows the reconstruction from Fig. 4(f) by eliminating with PCA. Some redundant background phases are also kept from the phase profiles, especially when the linear and spherical

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phases are no longer the dominant in total phase $\varphi(x, y)$. Figures 4(h) and 4(i) show the phase compensation using conventional fitting and the ADAF method based on the Zernike polynomials from the 1st term to the 15th term, respectively. Figure 4(j) shows the reconstructed wrapped phase aberration from Fig. 4(f) by using the ADAF method. The average calculating time is 5.63 s by the ADAF method. The PCA and fitting method incompletely compensate all the distorted regions in the phase distribution. But the aberration is almost canceled by using the ADAF method. The whole motivation is to ensure proper cell phase visualization for further analysis without a phase offset error. Thus, ensuring a flat phase in the background is crucial for correct analysis.

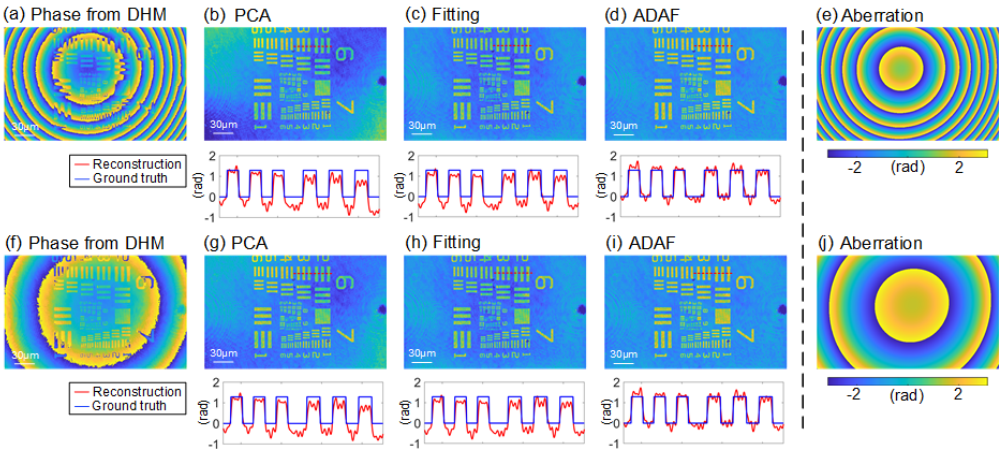


Fig. 4. The aberrations extraction under strong background and weak background by using a different method. (a) The reconstructed phase of the quantitative phase USAF1951 target from the hologram when spherical phase aberration dominates the principal components of the background phase. (b) The reconstruction from (a) by eliminating aberration with PCA. (c) The reconstruction from (a) by eliminating aberration with the conventional 2D surface fitting method. (d) The reconstruction from (a) by eliminating aberration with the ADAF method. (e) The reconstructed wrapped phase aberration by using ADAF method from (a). (f) The reconstructed phase of the quantitative phase USAF1951 target with weaker background aberration. (g) The reconstruction from (f) by eliminating aberration with PCA. (h) The reconstruction from (f) by eliminating aberration with the conventional 2D surface fitting method. (i) The reconstruction from (f) by eliminating aberration with the ADAF method. The phase profiles are shown below the reconstructions, where the blue curves are the ideal distribution, and the red curves are the reconstructed distributions. (j) The reconstructed wrapped phase aberration by using ADAF method from (f).

Figure 5 shows the eliminating aberration by using the ADAF method with Zernike basic functions. Three samples are considered phase USAF1951, label-free normal colon tissue slice, and label-free normal colonic mucosa. Figures 5(a1)-5(a3) show the reconstructed phase of three samples, respectively. In Figs. 5(a1)–5(a2), the NA and the magnification are 0.4 and 32×, respectively, which are the same as the case in Fig. 4. In Fig. 5(a3), the NA and the magnification are 0.8 and 20×, respectively, and the number of pixels is 2900×3800 with pixel's size 2.74μm. The tilt aberration becomes the leader role in Figs. 5(a1) and 5(a2) and some high-order aberration can still be seen. The parabolic and high-order aberration become obvious in Fig. 5(a3). For the structure like phase USAF1951 target, removal of the notorious aberration in phase imaging benefits the quantitative measurement of the object. Figures 5(b1)–5(b3) show the reconstructed phase after aberration compensation based on the Zernike polynomials. For the weak phase object in Fig. 5(b2), fibrous structures become evident by comparing with Fig. 5(a2) only when

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the aberration is corrected. Figures 5(c1)-5(c3) show the pseudo-3D view of reconstructed aberration by using the ADAF method based on the Zernike polynomials from the 1st term to the 15th term, respectively. The corresponding coefficients are reconstructed as shown in Figs. 5(d1)–5(d3). As shown in Figs. 5(a3) and 5(c3), the complex aberration can be extracted by the ADAF method for the typical tilt, astigmatism, and parabolic aberrations. The fitting model to higher order is expanded straightforwardly for more complex aberrations if high-order aberrations in the experiment must be removed. The average calculating time is 6.54 s by the ADAF method.

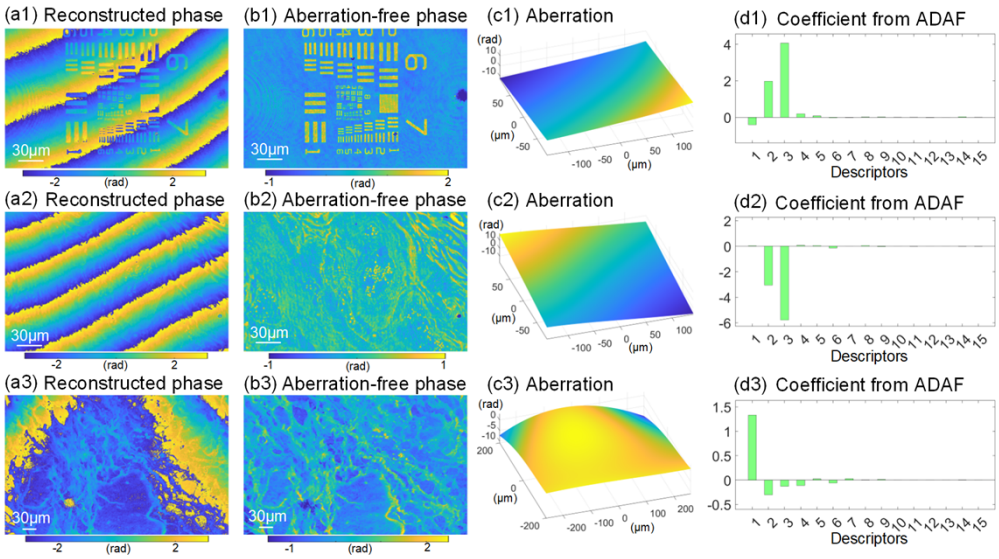


Fig. 5. The aberrations extraction by using the ADAF method based on the Zernike polynomials. (a1-a3) The reconstructed phase of the phase USAF1951, label-free normal colon tissue slice, and label-free normal colonic mucosa from the holographic phase microscopy, respectively. (b1-b3) The phase fields after aberration compensation of (a1-a3) from the first to the 15th term. (c1-c3) The pseudo-3D view unwrapped aberration fields from (a1-a3). (d1-d3) The reconstructed Zernike coefficients in (c1-c3) from the first to the 15th order polynomial.

Figure 6 shows the aberration elimination by using the ADAF method with standard basic functions. The raw total phases are the same as Figs. 5(a1)–5(a3). Figures 6(a1)–6(a3) show the reconstructed phase after aberration compensation based on the standard polynomials. Figures 6(b1)–6(b3) show the profiles comparison of the aberration-free phase based on standard and Zernike polynomials from Figs. 6(a1)–6(a3) and Figs. 5(b1)–5(b3), respectively, and the profiles of the objects come from the same cross-sectional lines in Figs. 6(a1)–6(a3). The red solid line and blue dotted line represent the reconstructions from Zernike and standard polynomials, respectively. The profiles of the aberration-free phase have high consistency whether the aberrations are extracted based on standard or Zernike polynomials by using the ADAF method. The corresponding coefficients of reconstructed aberration based on the standard polynomials from the 1st term to the 15th term are reconstructed, as shown in Figs. 6(c1)–6(c3). When the coefficients of high-order aberration are small, the aberration-free reconstruction can be compatible with each other whether the Zernike polynomials or standard polynomials are used in the ADAF method. The higher order can be expanded straightforwardly if high-order aberrations in the experiment must be removed. The average calculating time is 12.26 s by the ADAF method. The convergence of this iterative framework is that the relative error value between two

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adjacent iterations is lower than 10^{-5}, which is redundant for experiments. The calculation can be faster by setting stable iterations in the experimental holographic imaging. Relying on the improvement of computing, the learning-based method by integrating the proposed alternating direction framework can achieve background eliminations more efficiently.

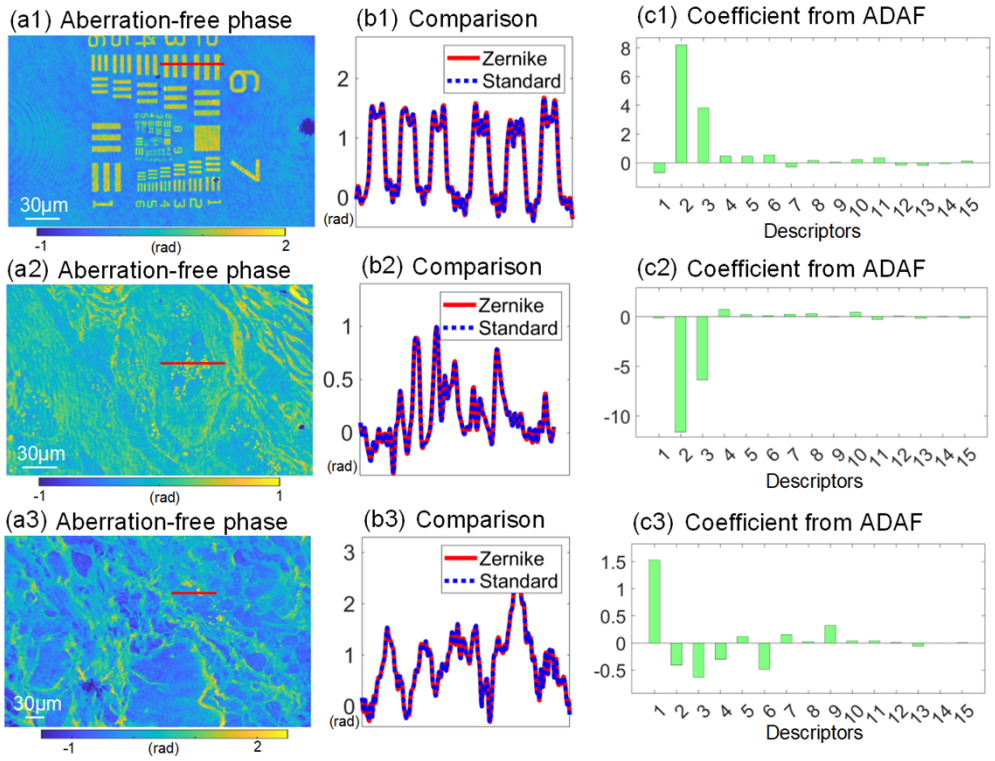


Fig. 6. The aberrations extraction by using the ADAF method based on the standard polynomials. (a1-a3) The phase fields after aberration compensation of Figs. 5(a1-a3) from the first to the 15th term. (b1-b3) The profiles comparison of aberration-free phase in the cross-sectional line from (a1-a3) and Figs. 5(b1-b3), respectively. The red solid line and blue dotted line represent the reconstructions from Zernike and standard polynomials, respectively. (c1-c3) The reconstructed standard coefficients in aberrations from the first to the 15th order standard polynomial.

3.3. *Phase aberration separation in a 3D sample*

We demonstrate that the ADAF method is compatible with a 3D sample. The M before L2 is replaced by the dual-axis galvano scanning mirror (GSM, GVS212, Thorlabs), which is located at a plane conjugate to the sample plane and the source illuminates a sample with various angles. The illumination beam is spirally scanned at different angles. The condenser lens is used with the same parameters as MO (UPlanXApo, 20×, NA = 0.8), as shown in Fig. 7(a). We used 130 incident angles with maximum illumination NA of 0.4. The sample is the polymethyl methacrylate (PMMA) bead with a diameter of 15 μm and its RI is about 1.49. The bead was immersed in oil ($n = 1.475$, sunflower seed oil) and sandwiched between two coverslips. Figure 7(b) shows the reconstructed amplitudes and phases from the hologram under different angles. Uneven background aberration is surrounded by the different angles of illumination. The Zernike polynomials from the zeroth to the 23rd order are considered for calculation. Figure 7(c) shows the reconstructed amplitudes and phases after eliminating phase aberrations under different

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angles. The RI tomogram is reconstructed based on the Rytov approximation [44–46]. Figure 7(d) shows the resultant RI tomogram from Fig. 7(b). Without aberration correction, the shape of the bead is distorted. It shows that diffraction tomography is vulnerable to background phase aberration in holographic reconstruction. Figure 7(e) shows the symmetric shape of the bead when the phase aberrations in the phase fields are corrected by the conventional fitting method. Figure 7(f) shows the symmetric shape of the bead when the phase aberrations in the phase fields are corrected by the ADAF method from Fig. 7(c). The result after the aberration correction shows good agreement with the ideal tomogram of a spherical bead. Figures 7(g) and 7(h) show the line profiles of the *x*-*y* and *x*-*z* cross-sections of the raw uncorrected tomogram, corrected tomogram by fitting method, and corrected tomogram by ADAF method, respectively. The phase aberration makes the final RI reconstruction deviate from the ideal value. The whole motivation is to ensure proper cell phase visualization for further analysis without a phase offset error. Thus, ensuring a flat phase in the background is crucial for accurate analysis.

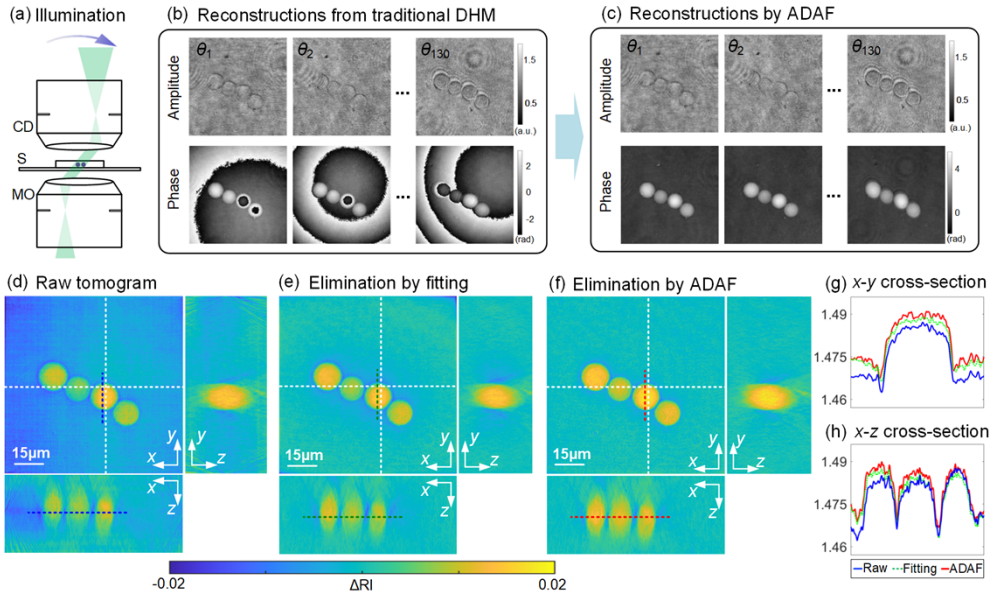


Fig. 7. Holographic imaging for a 3D sample. (a) Schematic representations of beam illumination from different angles at the objective back focal plane and sample planes for wide-field microscopy. (b) The reconstructed amplitudes and phases from the hologram under different angles. (c) The reconstructed amplitudes and phases after eliminating phase aberrations under different angles by using the ADAF method. (d) The reconstructed RI tomogram of the PMMA bead using raw complex amplitude images. (e) The RI tomogram after correcting the aberration by the conventional fitting method. (f) The RI tomogram after correcting the aberration by the ADAF method. (g) The line profiles of the *x*-*y* cross-sections of the raw uncorrected tomogram, corrected tomogram by conventional fitting method, and corrected tomogram by ADAF method. (h) The line profiles of the *x*-*z* cross-sections of the raw uncorrected tomogram, corrected tomogram by conventional fitting method, and corrected tomogram by ADAF method.

4. Conclusion

Interpretation of phase provides highly sensitive measurements and important novel parameters for studying biological physiological processes and material characterizing. We proposed a variable sparse splitting framework on aberration extraction based on the ADAF method. The

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optimization and regularization in phase are decomposed into two solutions with object terms and aberration terms. By formulating the aberration extraction as a convex quadratic problem, the background aberration can be fast and directly decomposed with the Zernike or standard polynomials. An accurate aberration surface is separated from the total reconstructed phase by imposing the sparsity regularization on the object and the coefficients of bases. The conjugate phase mask of the aberration is constructed to offset the phase aberration. Different samples such as phase resolution chart, label-free normal colon tissue slice, label-free normal colonic mucosa, and PMMA beads can be visualized in the quantitative phase and RI reconstruction with eliminating aberration. We demonstrate aberration-free 2D and 3D label-free imaging by correcting phase fields. It may serve as a vital tool to explore the function of overall label-free tissue and cell. We suggest that this method broadens the applications of quantitative phase imaging in some challenging conditions, and facilitates intelligent manufacturing advances by bringing various micro-, nano-, and meta-optical components into the practical scenes.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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